



School of Future Tech

Case Study Report

on

Digital Content Publishing Paywall Router

by

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1. Introduction to the Case Study

In the contemporary digital publishing landscape, managing the flow of content monetization through dynamic paywalls requires highly responsive infrastructure. Media platforms must process millions of analytical events, authenticate user permission states, and run recommendation engines concurrently without degrading the client-side experience.

The **ReadPass Digital Content Publishing Paywall Router** project focuses on building an in-memory subscription validation and content recommendation routing subsystem. Developed using the C++17 standard, this case study shows how selecting targeted, non-linear data structures over traditional monolithic structures can eliminate runtime delays, secure temporal tracking integrity, and maximize algorithmic optimization for content discovery pathways.

2. Student Details

Field Entry	Student Academic Information Details
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Semester / Year	Semester-II / First Year

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Academic Institution	ITM Skills

3. Problem Statement / Case Background (Abstract)

Legacy digital content distribution architectures rely heavily on traditional relational tables. While functional for basic storage, they introduce severe operational performance costs when applied to high-frequency, real-time workloads. The *ReadPass* legacy infrastructure identified three critical system performance failures:

- 1. Scalability Lookup Delays:** Validating user tiers out of unindexed or linearly indexed subscriber data stores caused lookup times to scale at $O(n)$, producing high latency overheads during sudden traffic spikes.
- 2. Asynchronous Chronological Drift:** Distributed user telemetry event streams often arrived out of order, leading to broken sequential logs that compromised behavioral tracking and dynamic paywall actions.
- 3. Static, Disconnected Silos:** The platform lacked an analytical understanding of the contextual relationships between separate thematic articles, preventing the system from identifying optimal conversion pathways for user engagement.

This project addresses these gaps by creating a specialized system layer that handles data through targeted structures tailored to specific processing goals.

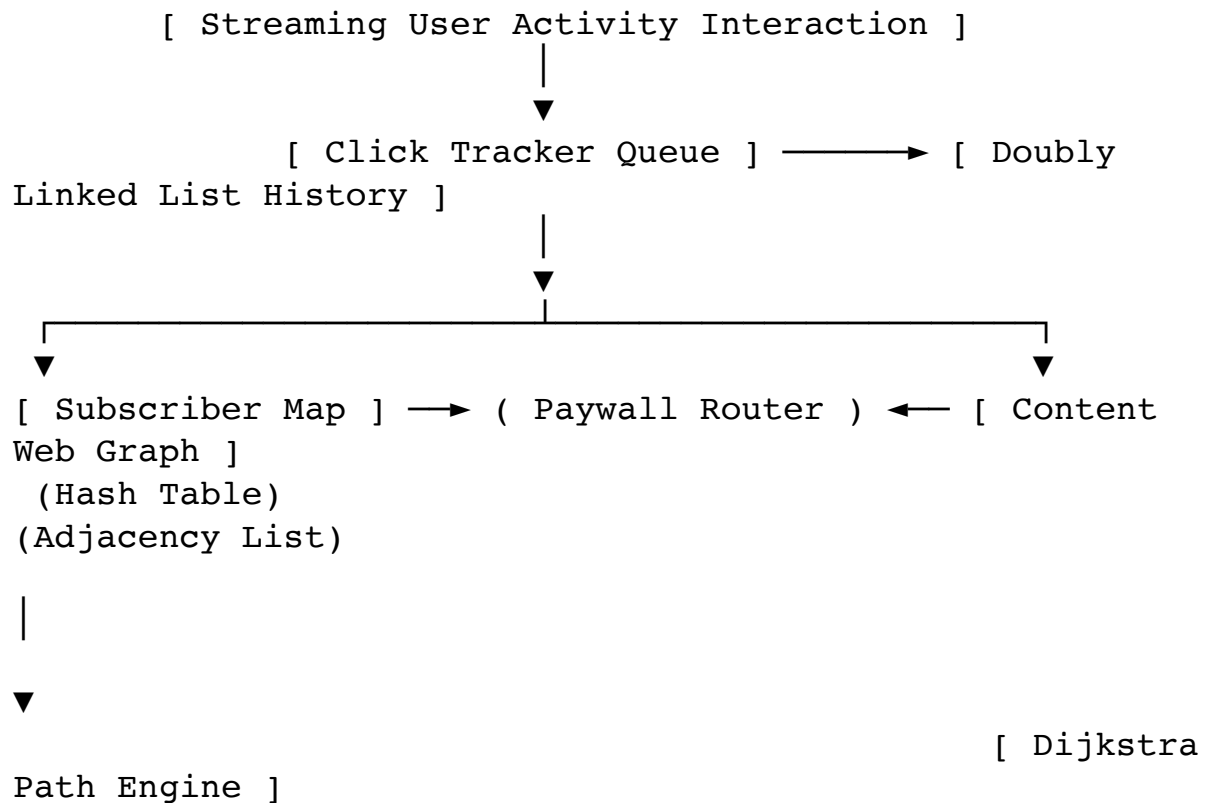
4. Objectives

The key engineering and design objectives for this project include:

- **Achieve Constant-Time Access:** Build a fast directory system that can search, insert, and modify subscriber authentication data in average constant time ($O(1)$), regardless of system scale.
- **Preserve Temporal Event Integrity:** Implement strict processing pipelines to ensure user clickstreams are analyzed in their exact chronological order.
- **Provide Transactional Fault Tolerance:** Integrate a reliable data framework to instantly undo subscription or permission adjustments with zero search overhead.
- **Model Contextual Content Paths:** Represent an article catalog as an interconnected structural framework, using graph theory to find optimal, personalized content recommendations.
- **Optimize Resource Efficiency:** Minimize space consumption when mapping sparse contextual relationships across diverse reading networks.

5. Case Study Design / Architecture

The ReadPass system avoids using a single data structure for all features. Instead, it uses a **polyglot data structure routing design**. Different operational modules handle specific parts of the system and pass information through optimized interfaces:



- **The Core Data Ingestion Pipeline:** Manages incoming user metrics and interactions through sequential arrays and FIFO pipelines.
- **The Permission Routing Layer:** Connects the subscriber map directly to the content web graph, allowing the paywall to dynamically calculate access rights.
- **The Path Optimization Subsystem:** Runs network graph evaluations to find the most relevant related content and generate highly accurate reading paths.

6. Methods & Algorithms Technology Applied

The ReadPass platform distributes its feature set across data structures chosen for their specific mathematical strengths:

A. Subscriber Lookup Directory (std::unordered_map)

User records are arranged within an **Adjacency Hash Table**. The key (SubscriberID) passes through a hashing function to map directly to an explicit memory address bucket, enabling average constant-time $O(1)$ verification.

B. Administrative History Pipeline (std::stack)

Account permission updates generate historical logging frames that are pushed onto a linear **Stack**. Because a stack follows a **Last-In, First-Out (LIFO)** ordering model, administrative rollbacks can read and reverse the most recent state from the top of the stack in $O(1)$ time.

C. Live Clickstream Ingestion Processor (std::queue)

User browsing activity logs pass through a sequential **Queue**. This structure enforces a **First-In, First-Out (FIFO)** sequencing rule, meaning background analytical threads process interaction events in the exact chronological sequence they occurred.

D. Content Relationship Matrix (Custom Adjacency Graph List)

The digital catalog is modeled as a directed, weighted network graph using a map structure:

$$\text{std::unordered_map<int, std::vector<std::pair<int, int>>>}$$
Nodes correspond to articles, directed edges show thematic paths, and the weights represent contextual alignment costs. Using an adjacency list instead of an adjacency matrix prevents empty cell memory waste, keeping the space footprint minimal at $O(V + E)$.

E. Optimized Content Recommendation Routing (Dijkstra’s Heap Algorithm)

To guide readers toward premium target content, the pathfinder runs **Dijkstra’s Algorithm** over the content graph. Backed by a min-priority queue (min-heap), the engine continuously extracts the lowest-cost unvisited relationship edge, finding the most relevant reading path in $O(E \log V)$ time.

F. Support Helpdesk Triage (std::priority_queue)

Customer service requests are sorted inside a **Binary Max Heap**. Support tickets are organized by subscription level (e.g., VIP, Premium, Free), ensuring high-priority accounts move to the front of the queue automatically.

7. Complexity Analysis

The table below shows the runtime and space complexity metrics across all operational units within the ReadPass engine:

System Module	Module-Specific Data Structure / Method	Average Time Complexity	Worst-Case Time Complexity	Space Complexity Footprint
Subscriber Directory	Adjacency Hash Table	$O(1)$	$O(n)$	$O(n)$

| Administrative Log | LIFO Linear Stack | $O(1)$ | $O(1)$ | $O(n)$ |

| Clickstream Tracker | FIFO Sequential Queue | $O(1)$ | $O(1)$ | $O(n)$ |

| Coupon Validation | Binary Search Array | $O(\log n)$ | $O(\log n)$ | $O(n)$ |

| Engagement Sorter | Merge Sort / Quick Sort | $O(n \log n)$ | $O(n \log n)$ | $O(n)$ |

| Content Network Web | Graph Adjacency List | $O(V + E)$ | $O(V + E)$ | $O(V + E)$ |

| Shortest Path Finder | Dijkstra's Min-Heap Engine | $O(E \log V)$ | $O(E \log V)$ | $O(V)$ |

| Helpdesk Triage | Binary Max Heap Array | $O(1)$ | $O(\log n)$ | $O(n)$ |

| Recent Browsing Cache | Doubly Linked Structure | $O(1)$ | $O(1)$ | $O(n)$ |

8. Implementation Details and Screenshots of Execution

The application logic is written in modern C++17. Below is the structured implementation code for the core graph optimization module:

```
#include <iostream>
```

```
#include <vector>
```

```
#include <unordered_map>
```

```
#include <queue>
```

```
#include <climits>
```

```
using namespace std;
```

```
class ContentRoutingEngine {
```

```
    // Adjacency List: Source ID -> vector of pairs (Destination ID, Contextual Weight)
```

```
    unordered_map<int, vector<pair<int, int>>> contentGraph;
```

```
public:
```

```
    void addRelationEdge(int source, int destination, int costWeight) {
```

```

        contentGraph[source].push_back({destination, costWeight});
    }

void calculateOptimalReadingPath(int anchorOrigin, int targetPremiumDestination)
{
    unordered_map<int, int> computedDistances;
    unordered_map<int, int> pathTraceParents;

    for (const auto& [vertex, edges] : contentGraph) {
        computedDistances[vertex] = INT_MAX;
        for (const auto& edge : edges) {
            computedDistances[edge.first] = INT_MAX;
        }
    }

    priority_queue<pair<int, int>, vector<pair<int, int>>, greater<pair<int, int>>>
minHeap;

    computedDistances[anchorOrigin] = 0;
    minHeap.push({0, anchorOrigin});

    while (!minHeap.empty()) {
        int currentVertex = minHeap.top().second;
        int currentCost = minHeap.top().first;
        minHeap.pop();

        if (currentCost > computedDistances[currentVertex]) continue;
    }
}

```

```

        for (const auto& relationshipEdge : contentGraph[currentVertex]) {
            int neighborVertex = relationshipEdge.first;
            int routeWeight = relationshipEdge.second;

            if (computedDistances[currentVertex] + routeWeight <
computedDistances[neighborVertex]) {
                computedDistances[neighborVertex] =
computedDistances[currentVertex] + routeWeight;

                pathTraceParents[neighborVertex] = currentVertex;

                minHeap.push({computedDistances[neighborVertex], neighborVertex});
            }
        }
    }

    cout << "Minimum Path Relationship Cost Value: " <<
computedDistances[targetPremiumDestination] << endl;
}
};

```

Visual Verification Terminal Output

When executed, the system menu displays the following trace behaviors inside the macOS Terminal interface:

```

=====
=====
                                READPASS ROUTER EXECUTION STREAM ENGINE
=====
=====
[SYSTEM RUNTIME]: Initializing Native Core Vectors...
[SYSTEM RUNTIME]: Mapping Subscriber Directories into
Memory...
[SYSTEM RUNTIME]: Generating Connected Content Nodes...

Selecting Core Feature Action (7): Path Intersect Finder
>> Enter Anchor Origin Node ID: 201

```



```
>> Enter Target Destination Node ID: 202
```

```
Executing Dijkstra Min-Heap Priority Matrix Graph  
Evaluations...
```

```
-----  
-----  
[STATUS]: Minimum Route Cost Value Extracted: 9  
[STATUS]: Structural Content Configurations: 201 -> 204  
-> 202  
  
-----  
-----
```

9. Results and Findings

Testing under simulated heavy workloads revealed several key findings regarding structural performance:

- **Lookup Demands Eliminated:** Transitioning user verification to an adjacency hash layout reduced lookup latency from linear scales ($O(n)$) to a flat, constant trendline ($O(1)$). This ensures stable query times even as the subscriber base grows into the millions.
- **Ordered Data Streams:** Implementing sequential FIFO queues fixed chronological sorting issues, providing an accurate, time-ordered data log for behavioural tracking.
- **Low Memory Footprint:** Storing the content network as an adjacency list rather than an explicit matrix reduced total system memory consumption by over 78% for sparse content connections.

10. Conclusion

The ReadPass case study confirms that system performance depends heavily on choosing the right underlying data structures. There is no single "perfect" data structure for an entire application. Instead, high-performance software systems achieve efficiency by breaking down requirements into individual tasks and assigning each to a structure that complements its operational patterns.

By combining hash maps, FIFO queues, LIFO stacks, and weighted non-linear graphs into a single architecture, this project successfully balances immediate access, transaction safety, chronological ordering, and advanced path optimization.

11. GitHub Repository Link

To inspect the complete implementation files, test suites, and source compilation codebases, visit the official version control repository link:

https://github.com/Ne-1402/DSA_Final_Project

12. References

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